

Lab

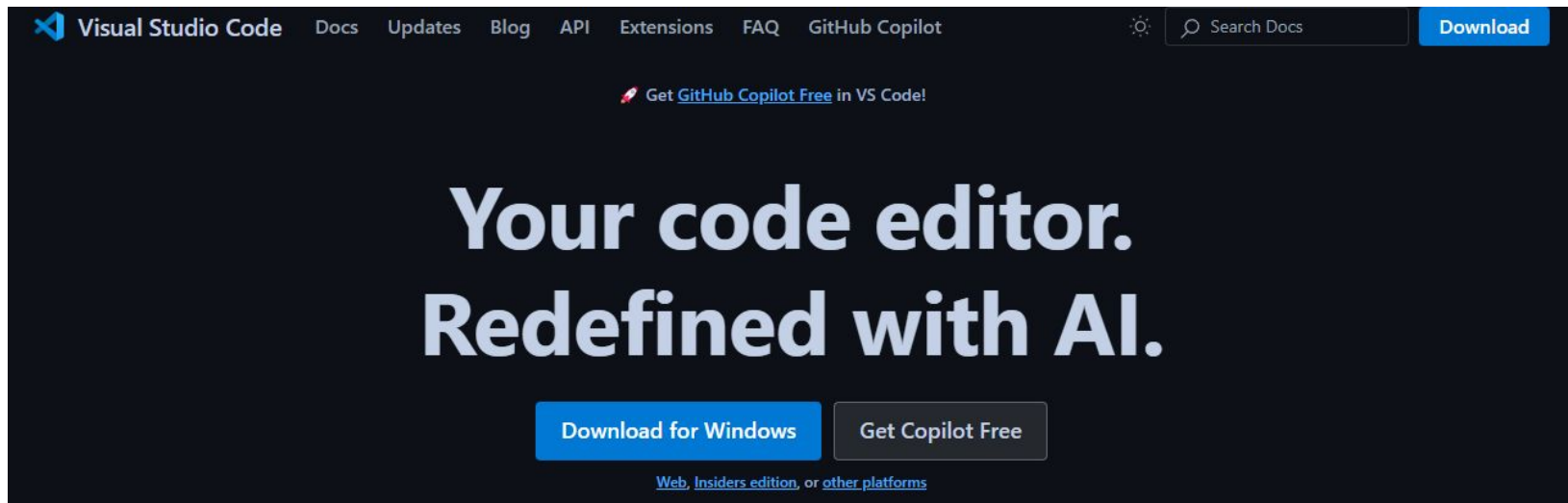
Lab Activities

1. Set up IDE - Visual Studio Code
2. Set up Git / GitHub
3. Set up Docker Container
 - a. DockerFile
 - b. DockerCompose
 - c. Requirements.txt
4. Run first py script
5. Run first notebook
6. Commit work

Lab

Set up IDE - Visual Studio Code

Download: <https://code.visualstudio.com/>

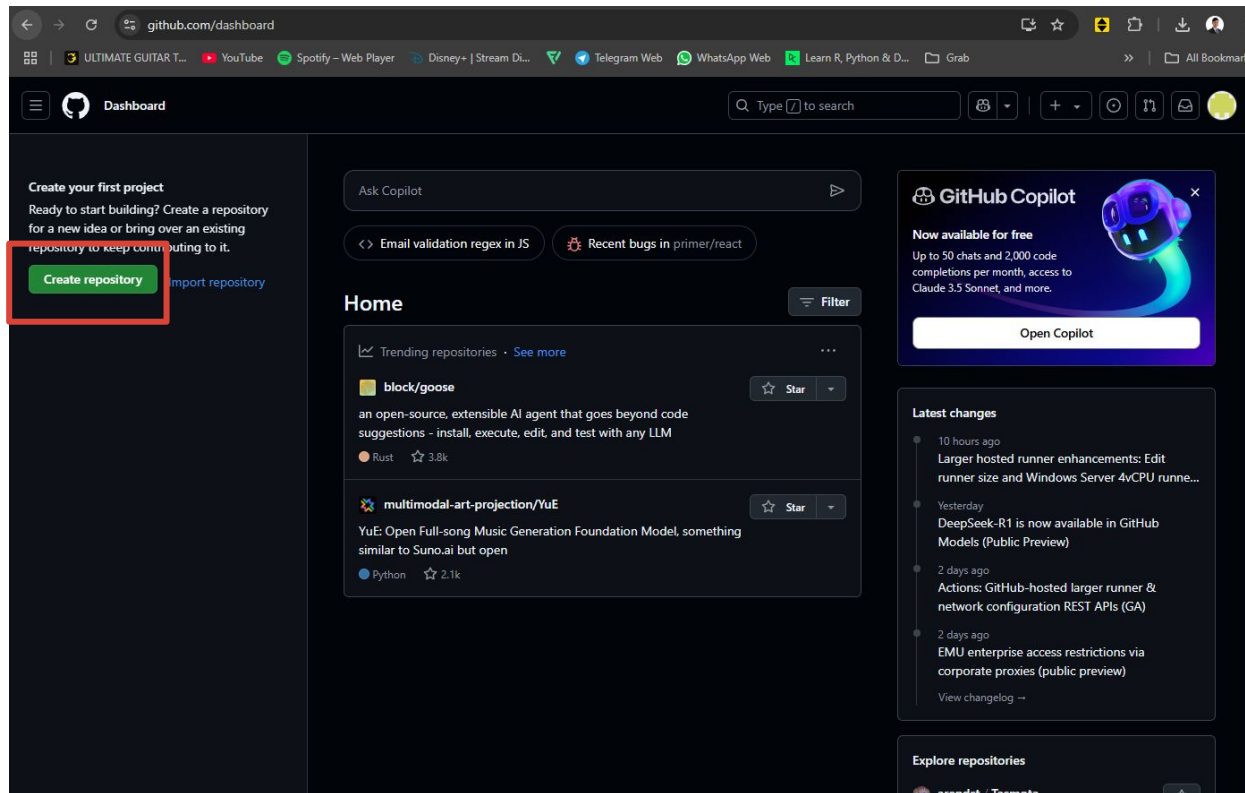


Lab

Set up Git / GitHub

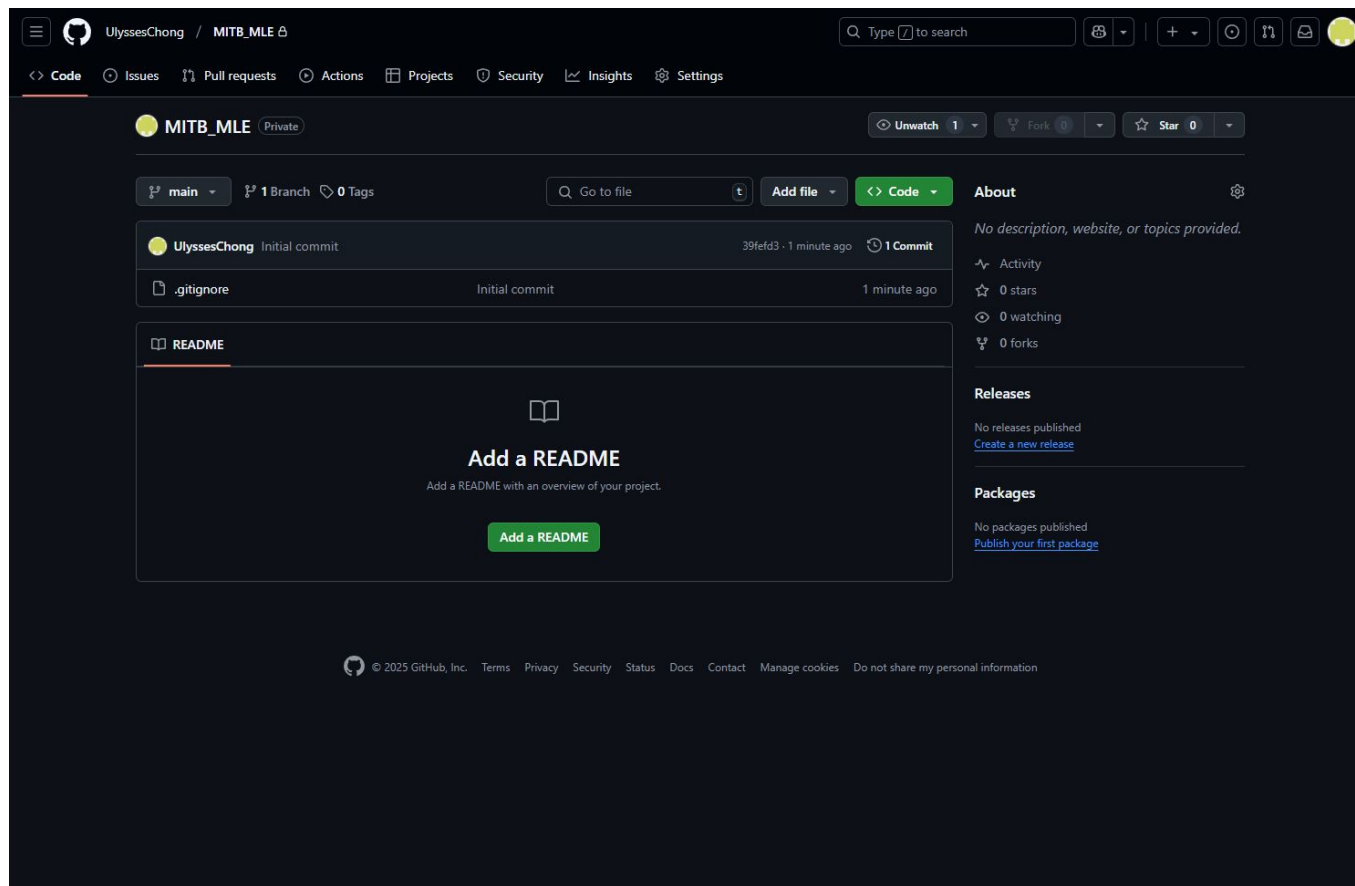
<https://github.com/>

highly recommend to enable 2FA



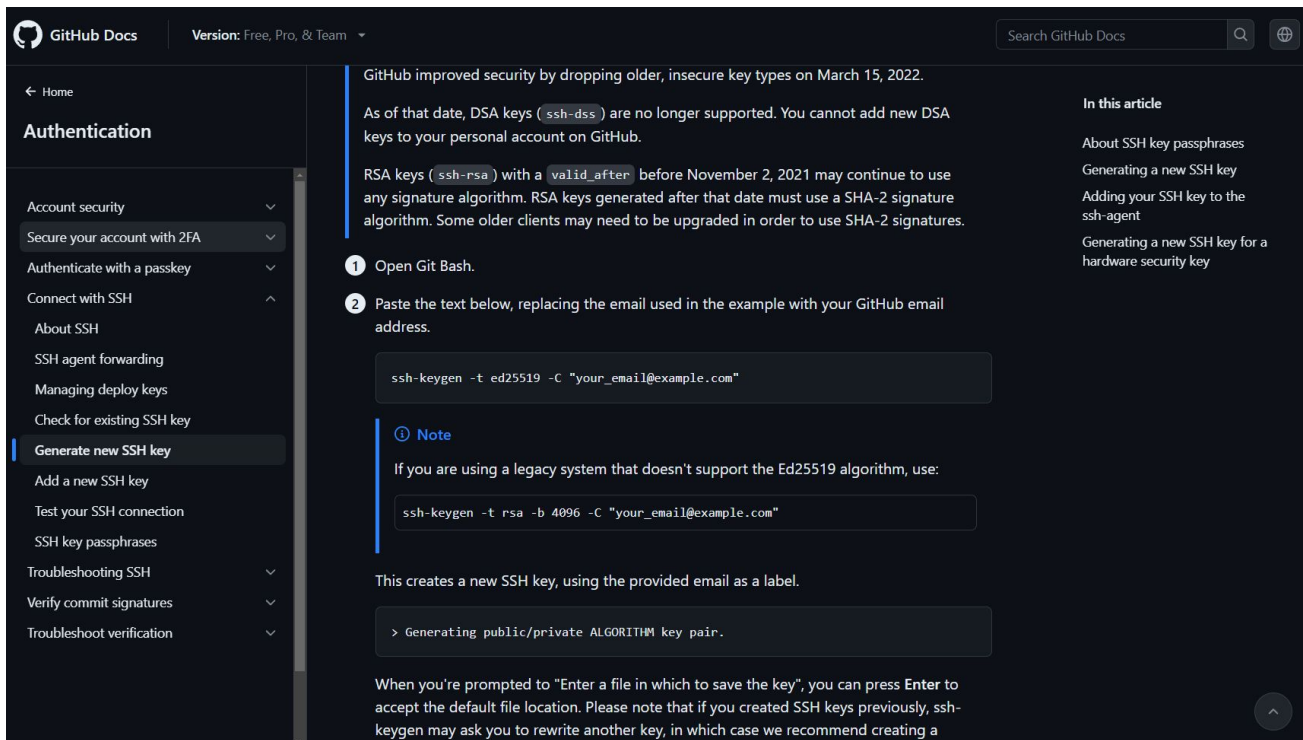
Lab

Create public repo Initialize .gitignore to Python



Set up SSH

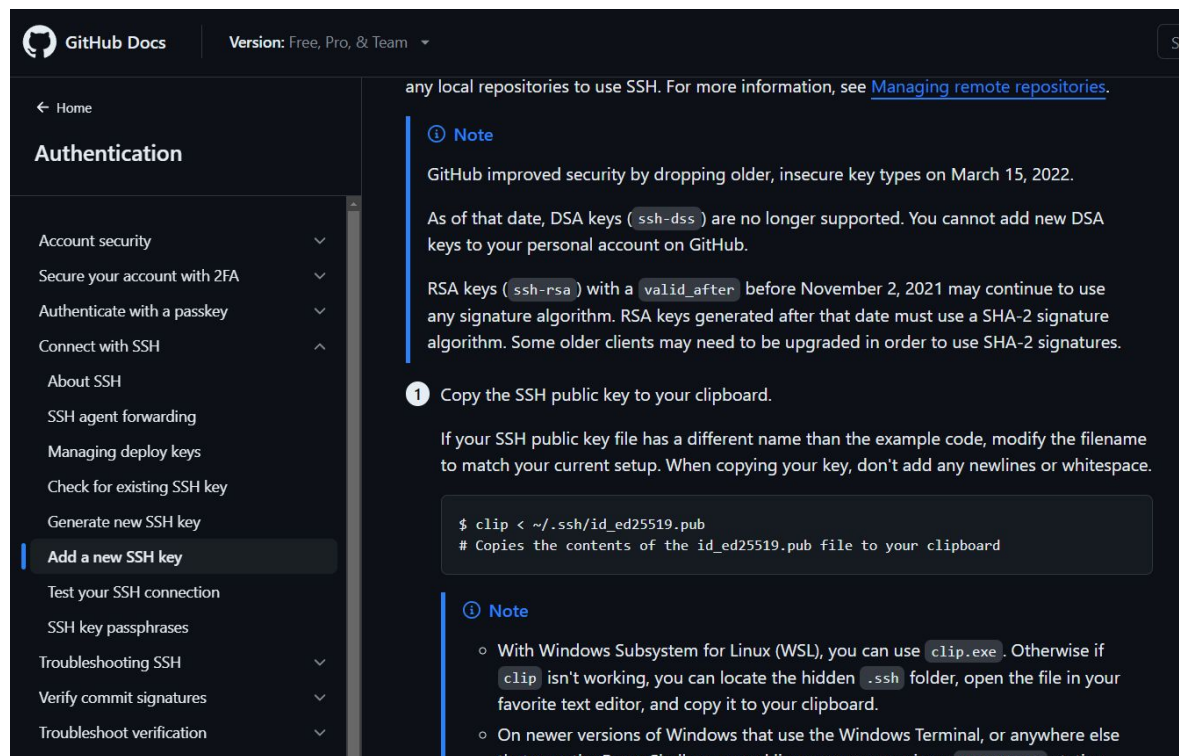
<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>



The screenshot shows the GitHub Docs website. The left sidebar is dark-themed and contains a navigation menu under the heading "Authentication". The menu items are: Account security, Secure your account with 2FA, Authenticate with a passkey, Connect with SSH, About SSH, SSH agent forwarding, Managing deploy keys, Check for existing SSH key, Generate new SSH key (highlighted with a blue bar), Add a new SSH key, Test your SSH connection, SSH key passphrases, Troubleshooting SSH, Verify commit signatures, and Troubleshoot verification. The main content area is white and features a header with "GitHub Docs", a version dropdown set to "Free, Pro, & Team", and a search bar. Below the header, a blue banner states: "GitHub improved security by dropping older, insecure key types on March 15, 2022. As of that date, DSA keys (ssh-dss) are no longer supported. You cannot add new DSA keys to your personal account on GitHub." The main text explains that RSA keys (ssh-rsa) with a valid_after date before November 2, 2021, may continue to use any signature algorithm, but keys generated after that date must use a SHA-2 signature algorithm. Two numbered steps are provided: 1. Open Git Bash. 2. Paste the text below, replacing the email used in the example with your GitHub email address. A code block shows the command: `ssh-keygen -t ed25519 -C "your_email@example.com"`. A note indicates that for legacy systems not supporting Ed25519, the command `ssh-keygen -t rsa -b 4096 -C "your_email@example.com"` should be used. Below the code, it states: "This creates a new SSH key, using the provided email as a label." A terminal snippet shows the prompt: `> Generating public/private ALGORITHM key pair.`. At the bottom, a note says: "When you're prompted to 'Enter a file in which to save the key', you can press Enter to accept the default file location. Please note that if you created SSH keys previously, ssh-keygen may ask you to rewrite another key, in which case we recommend creating a". The right sidebar, titled "In this article", lists links: About SSH key passphrases, Generating a new SSH key, Adding your SSH key to the ssh-agent, and Generating a new SSH key for a hardware security key.

Set up SSH

<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>



The screenshot shows the GitHub Docs website. The left sidebar is dark-themed and contains a navigation menu under the heading 'Authentication'. The menu items are: Account security, Secure your account with 2FA, Authenticate with a passkey, Connect with SSH, About SSH, SSH agent forwarding, Managing deploy keys, Check for existing SSH key, Generate new SSH key, Add a new SSH key (highlighted with a blue bar), Test your SSH connection, SSH key passphrases, Troubleshooting SSH, Verify commit signatures, and Troubleshoot verification. The main content area is light gray and shows the 'Add a new SSH key' page. It includes a 'Note' section stating that GitHub improved security by dropping older, insecure key types on March 15, 2022, and that DSA keys are no longer supported. It also mentions that RSA keys generated after that date must use a SHA-2 signature algorithm. Below the note, there is a numbered list starting with '1 Copy the SSH public key to your clipboard.' followed by instructions to modify the filename if necessary. A code block shows the command: `$ clip < ~/.ssh/id_ed25519.pub` with a comment: `# Copies the contents of the id_ed25519.pub file to your clipboard`. Another 'Note' section at the bottom provides instructions for Windows users, mentioning WSL and the `clip.exe` command, and for newer versions of Windows using the Windows Terminal.

GitHub Docs Version: Free, Pro, & Team

← Home

Authentication

- Account security
- Secure your account with 2FA
- Authenticate with a passkey
- Connect with SSH
- About SSH
- SSH agent forwarding
- Managing deploy keys
- Check for existing SSH key
- Generate new SSH key
- Add a new SSH key**
- Test your SSH connection
- SSH key passphrases
- Troubleshooting SSH
- Verify commit signatures
- Troubleshoot verification

any local repositories to use SSH. For more information, see [Managing remote repositories](#).

Note

GitHub improved security by dropping older, insecure key types on March 15, 2022. As of that date, DSA keys (`ssh-dss`) are no longer supported. You cannot add new DSA keys to your personal account on GitHub.

RSA keys (`ssh-rsa`) with a `valid_after` before November 2, 2021 may continue to use any signature algorithm. RSA keys generated after that date must use a SHA-2 signature algorithm. Some older clients may need to be upgraded in order to use SHA-2 signatures.

1 Copy the SSH public key to your clipboard.

If your SSH public key file has a different name than the example code, modify the filename to match your current setup. When copying your key, don't add any newlines or whitespace.

```
$ clip < ~/.ssh/id_ed25519.pub
# Copies the contents of the id_ed25519.pub file to your clipboard
```

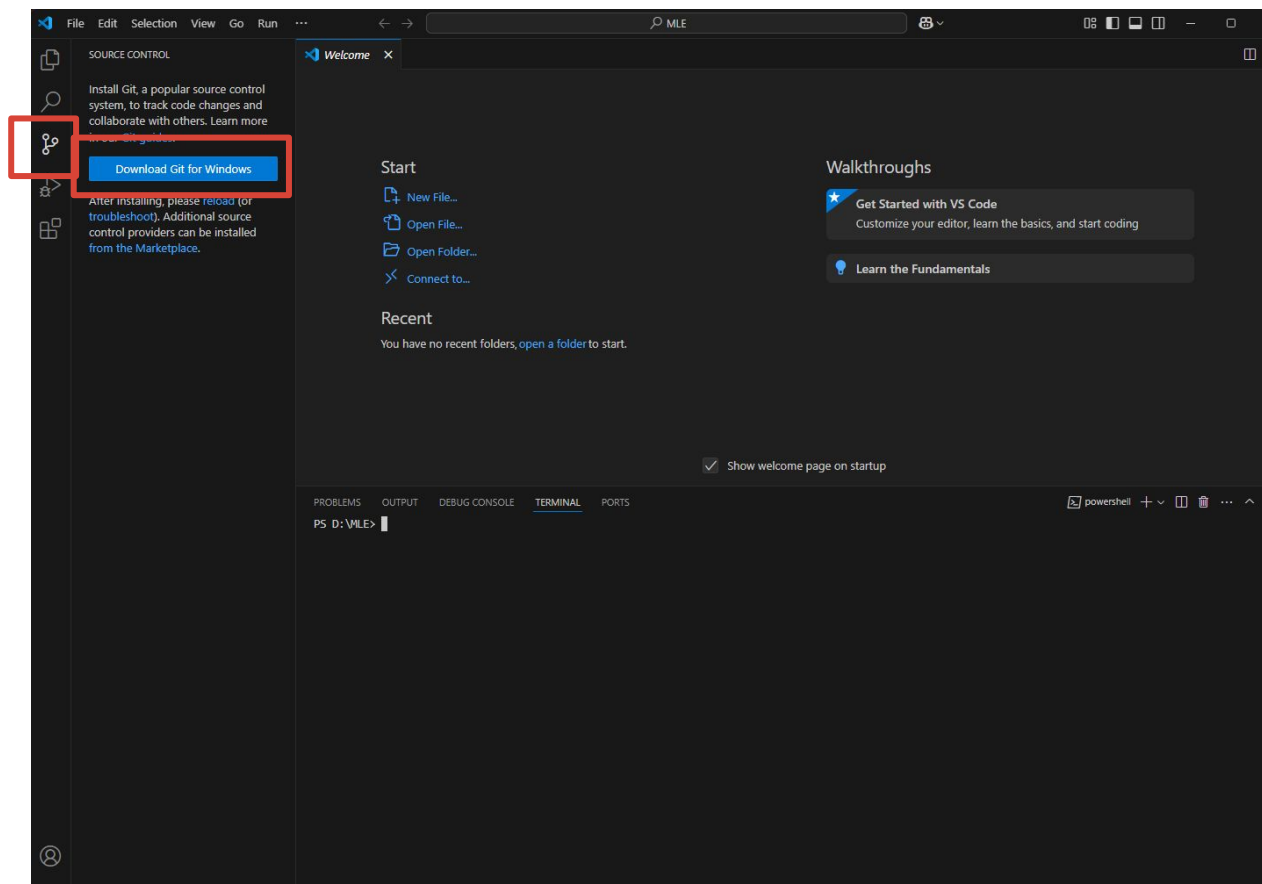
Note

- With Windows Subsystem for Linux (WSL), you can use `clip.exe`. Otherwise if `clip` isn't working, you can locate the hidden `.ssh` folder, open the file in your favorite text editor, and copy it to your clipboard.
- On newer versions of Windows that use the Windows Terminal, or anywhere else that uses the PowerShell command line, you may receive a `PowerPrompt` stating

Lab

set up git for windows

<https://git-scm.com/downloads/win>



Lab

clone your repo
git clone ...

Lab

quick git cmd:

git checkout main

git pull

git checkout -b "branch-1" # use any branch name with -
[make your changes to code]

git status

git add .

git commit -m "commit message"

git status

git push

Set up Docker Container

<https://www.docker.com/>

The screenshot shows the Docker website homepage. The main headline is "Develop faster. Run anywhere." with the subtext "Build with the #1 most-used developer tool". Below this, there are buttons for "Download Docker Desktop" and "Learn more about Docker". A dropdown menu is open from the "Download Docker Desktop" button, showing options for different operating systems and architectures: "Download for Mac - Intel Chip", "Download for Mac - Apple Silicon", "Download for Windows - AMD64", "Download for Windows - ARM64 (BETA)", and "Download for Linux". In the background, a Docker Desktop interface is visible, showing a list of containers. The interface includes a sidebar with navigation options like Containers, Images, Volumes, Dev Environments, Docker Scout, and Learning Center. The main area displays a table of running containers.

Name	Image	Status	CPU (%)	Ports	Actions
welcome-to-docker	docker/welcome-to-docker	Running	1.43%	52.52	[Play] [Stop] [Refresh]
nginx	nginx:latest	Running	0%	80.80	[Play] [Stop] [Refresh]

At the bottom of the page, there is a cookie consent banner with buttons for "Cookies Settings", "Reject All", and "Accept All Cookies".

Lab

docker-compose build

docker-compose up

docker-compose down

Lab

Lab 1 completed sample

The screenshot shows a GitHub repository interface for the user UlyssesChong, specifically the MITB_MLE repository. The left sidebar displays the file explorer for the 'main' branch, with the 'lab_1' directory expanded. The files listed in 'lab_1' are DockerFile, docker-compose.yaml, hello.ipynb, hello.py, requirements.txt, .gitignore, and playground.py. The main content area shows the commit history for the 'lab_1' directory. A commit by UlyssesChong titled 'specify package versions' is highlighted. Below this, a table lists the files and their corresponding commit messages.

Name	Last commit message
..	
DockerFile	set up dockerfile
docker-compose.yaml	set up dockerfile
hello.ipynb	run docker
hello.py	run docker
requirements.txt	specify package versions

Are you feeling like a programmer
now?



An aerial night photograph of Singapore. The foreground shows a large, modern building with a brown roof and glass facade, identified by the 'SMU' logo. The background features a dense city skyline with numerous illuminated skyscrapers under a dark sky. The text 'Questions?' is overlaid in a bold, orange font on the left side of the image.

Questions?

Thank You